Cor Coffee Spring Syrup Analysis Feb - Mar 2026

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Introduction

Cor Coffee is a non-profit, catholic coffee shop located in the St John Paul II Newman Center on South Morgan St in Chicago Illinois with the mission statement “Helping college students encounter Christ and be fully alive in him”. Using the the sales data given to me I was able to run an analysis on their new syrups that they launched for the spring. The new syrups include: Ube, that was launched in February, Lavender which has been released every march since 2022, and Pistachio, that was newly released in march alongside Lavender. The questions that I will answer in this analysis are:

1. What syrup is the most popular with the drinks?
2. When does Ube, Lavender, and Pistachio sell the most?
3. What milk goes best with the syrups?
4. What is the difference in sales of the syrups between Hot vs Iced?
5. How many people from previous years are repeat drinkers of Lavender?
6. How much do spring syrups contribute to the total revenue?
7. Do customers that order spring syrups tend to be a different kind of customers?
8. What predicts whether a customer orders Ube, Lavender, or Pistachio?

Data

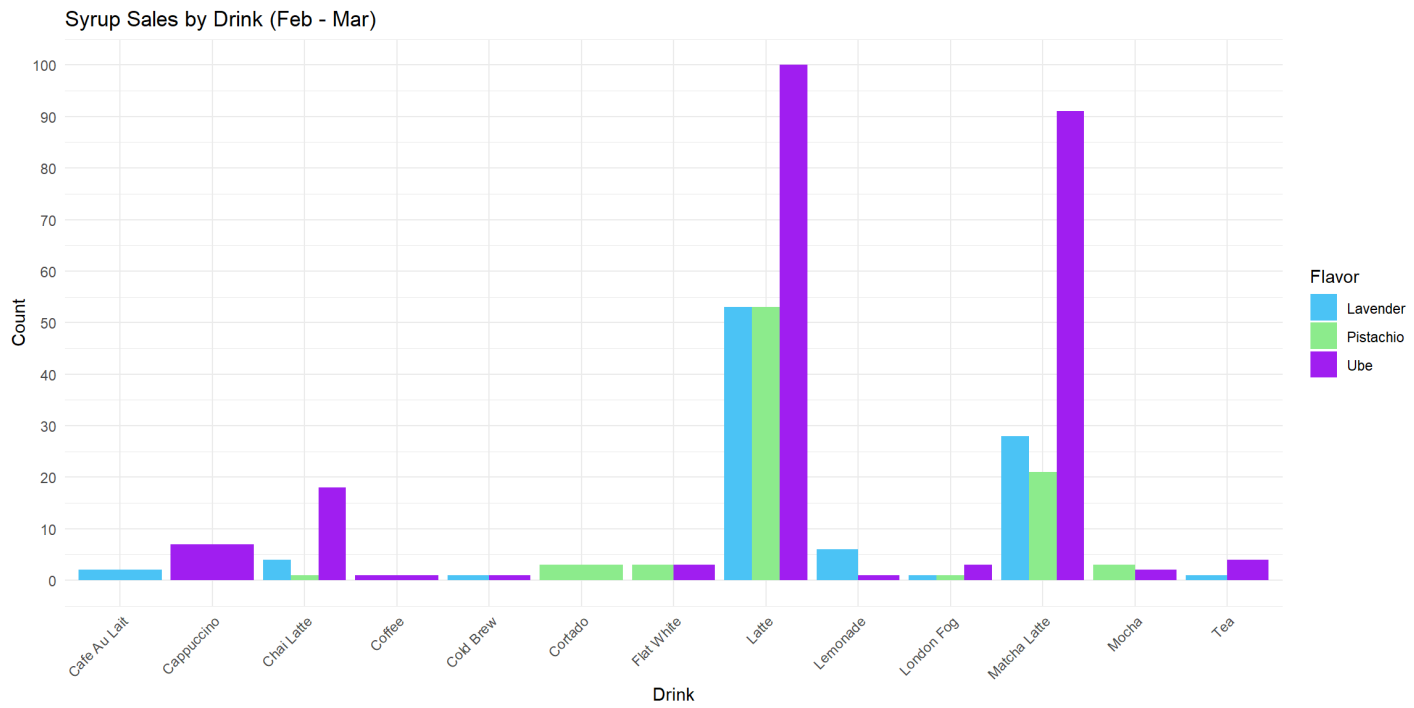
The data was obtained from the app *Square*, a comprehensive point-of-sale system and payment that is designed to support businesses. *Square* has a dashboard that contains all of the data that was needed. This data set had roughly 3002 observations. The variables that were used consistently throughout this project was the Modifiers.Applied and Item, since it contained all the drinks, syrups, milks, and the temperature of the drinks used and the function `grepl()` was used to differentiate the key words needed. Other variables used were Net.Sales, however this variable was of a character type and in the form of ($#.##) and I needed to clean it by using `substr()` to exclude the the “$” sign and then use the function `as.numeric()` to convert it a number. I used a similar process for the variable Time and Date where I got rid of any non-numeric characters and then changed it to a number. For Time specifically, I only kept the hour since the exact minutes didn’t make a big difference, while for Date I got both the days and the month.

Finally, for the question about sales from previous years, there needed to be a reliable variable that tracked the same customers. There were three variables to choose from, Customer.ID, Customer.Reference.ID, and PAN.Suffix. Customer.ID contained 2189 empty observations, Customer.Reference.ID contained 3000 empty observations, and Pan.Suffix contained 174 empty observations. It was decided to use PAN.Suffix since it contained the least amount of empty observations. PAN.suffix is just the last 4 digits of a customer’s card number that is saved onto *Square* and this is a good metric of comparing customers since they can be identified by the same PAN Suffix and Cor Coffee is cashless, so the only way to pay is through cards and gift cards. There are only 2 downsides: We lose about 6% of data but PAN.Suffix is our best options since it still contains 94% of all

observations in the data set, but more importantly customers could use a second card (or gift card) and this would mess up the data. Every other variable in the data set was discarded since they weren't used It is also important to note that Cor is usually open on days the University of Illinois Chicago (UIC) is open, from 9 Am - 4 pm (With a break from 11:00 - 12:35 pm for Mass). This data is from February 1st - March 20th 2026 so the semester is technically not over yet, however, there are only about 5-6 weeks left so the data won't change drastically and the conclusions will be about the same.

What syrup is the most popular with the drinks?

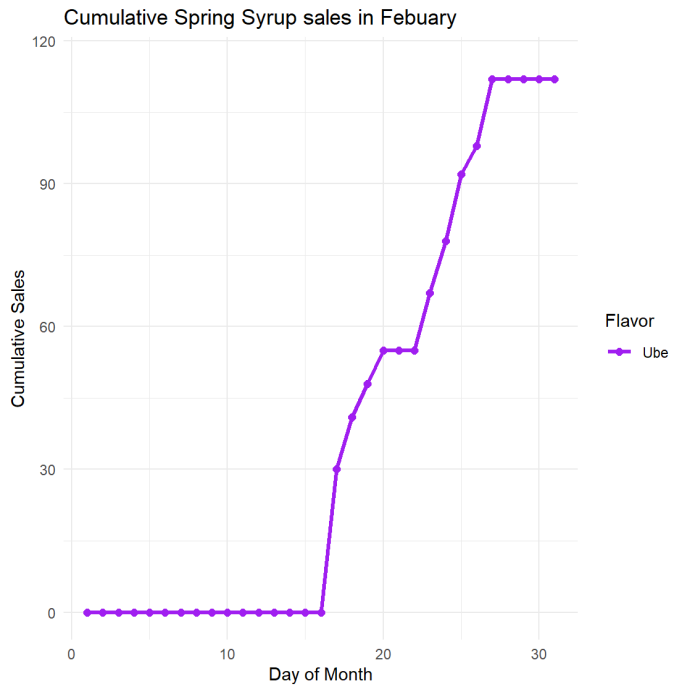
By grouping together the types of drinks with the syrups customers got, filtering out non-drink Items, and drinks that have never gotten a syrup (such as a Breve), and plotting the counts, a graph showing the results was produced:

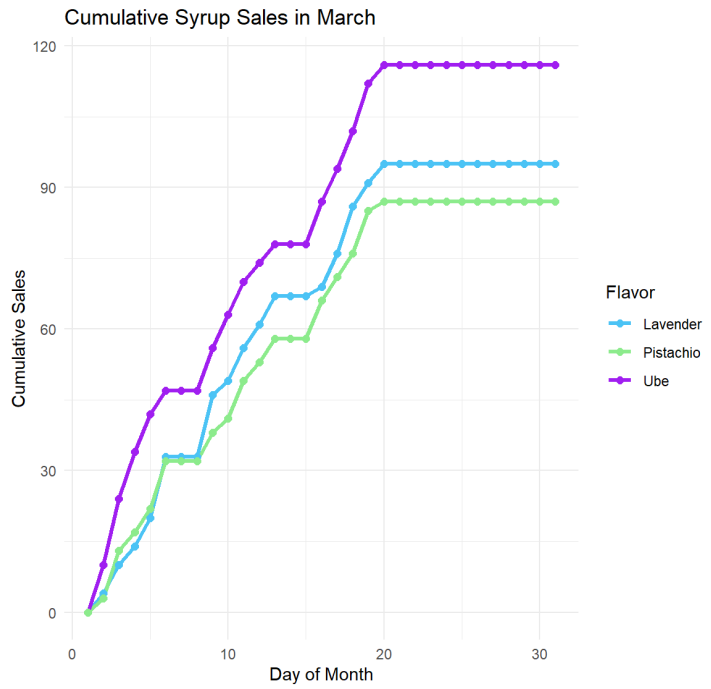


It is very apparent that Lattes and Matchas are the two drinks with the highest syrup count by a very large margin, with the Chai Latte having the third highest syrup count. In all three of the drinks it can be seen that the Ube has the most orders. However, Ube came out about a month earlier in advance so it has had more opportunities to be ordered compared to the Lavender and Pistachio. The Lavender and the pistachio are pretty close in terms of order count, however Lavender does have a noticeable edge. Ube has been ordered in 11 drinks, while Lavender has been ordered in 8 drinks, and Pistachio has been ordered in 7 drinks. This shows that Ube is not only more popular, but also is able to be used in a wider variety of drinks.

When does Ube, Lavender, and Pistachio sell the most?

By getting the days of all the orders and finding the syrups associated with those orders, we're able to find the sum of each syrup on a day. Finally we get every syrup's cumulative sum from when the syrup came out to March 20th (when the data was collected). This is the resulting graph:





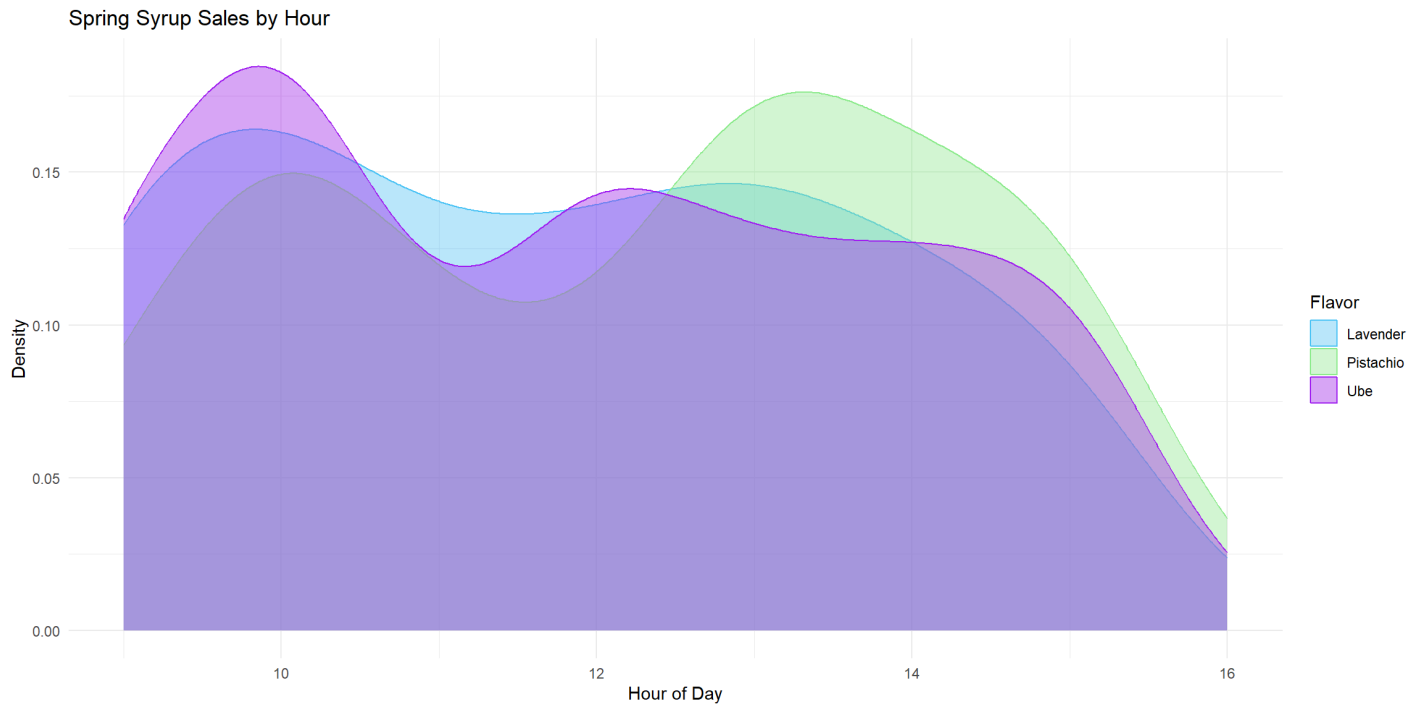
The graph on the left is the graph for cumulative sales of just the Ube since it came out a month early than the other two, while the graph on the right is the graph for the cumulative sale of all three. Looking at the February graph, we can see a sudden spike from the 17th. The Ube came out on the 17th of February and it seems as though the syrup being completely new is what drove a sudden spike in sales. From one day, we managed to get 30 orders which is the biggest growth we've seen from any of the spring syrups. Looking at either graphs, we don't ever get spike like that again from any of the syrups. After that first day, the Ube grows at more of a steady rate, but it still grows much faster than the Lavender and the Pistachio.

What is interesting is that Lavender isn't as popular as Ube despite being released every March so students are familiar with it. Instead Lavender grows at more of a consistent and steady rate rather than an exponential one, even on the day that it came out. Pistachio is equally as intriguing. It starts with a growth greater than Lavender for about the first 3 days, but Lavender then overtakes the sales of Pistachio and continues to out sell it. There are multiple points were the graph flat lines, this is due to Cor being closed on the weekends and closed for spring break (after march 20th).

A possible explanation for syrup's growth could be that the more new and versatile a syrup is, the more growth it will get. This can seen with the new flavors Ube and Pistachio that were both new this year and both had really good sales at the beginning, while Lavender has been sold for the past 3 years, so people are already familar with it. But I think the reason why Ube has such high growth in march is because it is much more versatile than Lavender and Pistachio, we saw from the Syrup sales by drink that the Ube had the most versatility, customers put it in their Lemonades, London Fogs, Teas, Chai, etc. Lavender is also versatile, as it had 8 different drinks, but since it's already well-known, people might be less inclined to try it.

What times do the syrups tend to get sold the most?

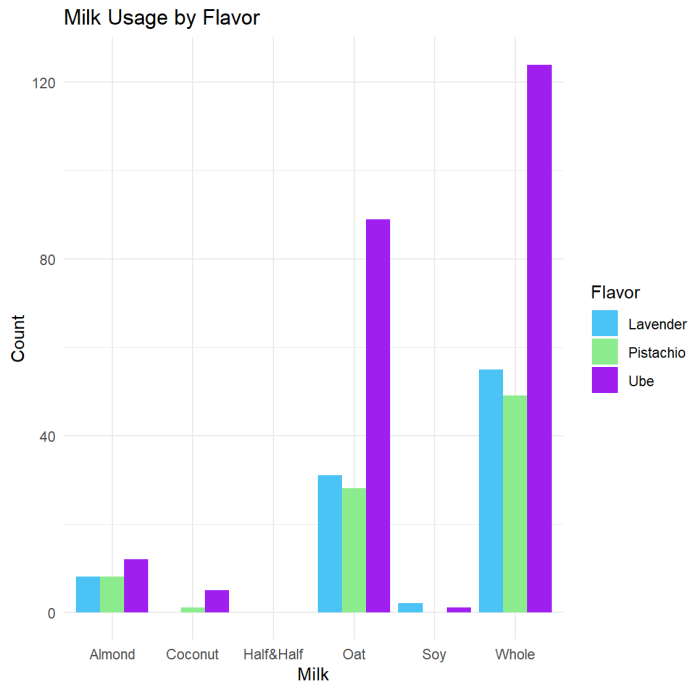
Using the Time variable from the data frame, we extracted the hour and turned it into a numeric value. Using this new value, we are able to gather all the spring syrup's order times and plot the data:

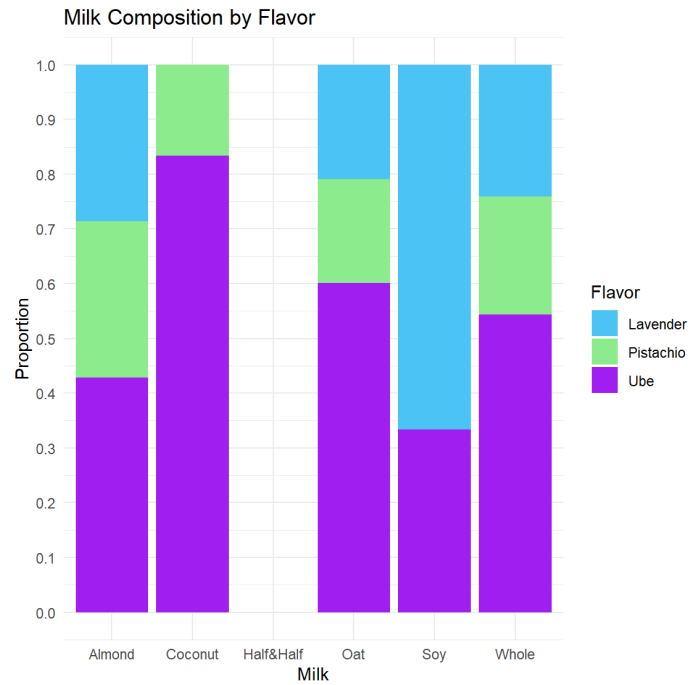


This is a density plot, where the height represents the frequency of sales. We can see that Ube sales peak around 9:30–10:00, roughly an hour after opening, with a smaller secondary peak around 12:30, which coincides with the store reopening after mass. After this secondary peak, the Ube sales start to slowly die down. In contrast, Lavender shows a more consistent sales pattern throughout the day, gradually tapering off after 2 PM, but having small peaks in sales around 9:30 AM and 1:00 PM. Pistachio starts with lower sales in the morning but gains momentum in the second half of the day, reaching levels comparable to Ube's morning peak, with the syrup also having a peak at around 9:30 AM. All the syrups also tend to have higher sales during the first hour of opening, which aligns with the morning rush, consistent with the store opening time and die down at roughly the same time which could coincide with when students leave campus since UIC is a 90% commuter school.

What milk goes best with the syrups?

By gathering different types of milks and syrups, a matrix of counts of orders with milk choice and syrup choice is created. There were some blanks for the milk type, this could be due to the baristas not selecting a milk when the customer specifies Whole milk since there is no extra charge. This was fixed by writing code that went through every observation and replaced blanks with Whole milk. Then a counts bar plot of syrups and milk types alongside a proportion graph of syrup preference in milk is created. Here are the graphs:





Whole and Oat have the most counts by a lot, which makes sense as Whole milk is the default milk and Oat milk tends to be the popular alternative around those that are lactose intolerant. Looking at the proportion graph, the Ube syrup dominates in flavor with the usages being about 43% in Almond, 82% in Coconut, 60% in Oat, 31% in Soy, and 55% in Whole. It seems to have a very high percentage in coconut, but that's more so because of a lack orders rather than just popularity. Lavender and Ube seem to have a more equal distribution with milk choice, with Lavender having slight edge in preference. This does make sense since Lavender and Pistachio are close in total sales as we saw from the cumulative sales graph.

To see if there really was a relationship between milk choice and syrup choice, a Chi-Square test of independence was run. The hypothesizes for the test are:

- **H_0:** There is no relationship between milk choice and syrup choice
- **H_1:** There is a relationship between milk choice and syrup choice

Set the significance level:

- **α:** 0.05

The Chi-Square test assumptions are listed below:

1. Both of the variables, milk choice and syrup choice, are categorical variables
2. Since a customer's order does not effect another customer's order, we can safely make the assumption that every observation is independent
3. Each observation does only fall into one cell i.e every observation has one milk type and one syrup type and not multiple
4. Random sampling is where things get tricky. Since all of the data was collected in this one coffee shop, it is harder to say that it was random, but for our purposes we only need it for Cor and we don't have control over our customers so there is some randomness.
5. Unfortunately, our Chi-Square test is not able to be run since our expected test condition is not met:

```
##                Ube   Lavender  Pistachio
## Whole   127.525424 52.9975787 47.4769976
## Oat      82.779661 34.4019370 30.8184019
## Almond   15.661017  6.5084746  5.8305085
## Soy       1.677966  0.6973366  0.6246973
## Coconut   3.355932  1.3946731  1.2493947
```

All of the expected counts of Soy and Coconut are less than 5 so we fail this assumption.

Fortunately, there is an alternative to the Chi-Square test, known as the Fisher's test that is used to see if there is a relationship between two categorical variables when the expected counts are low.

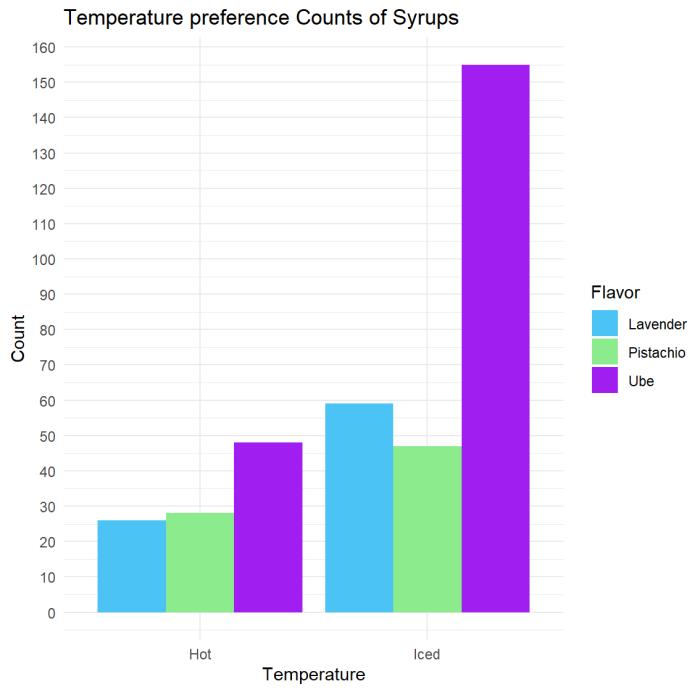
```
##
##  Fisher's Exact Test for Count Data
##
## data:  milksclean
## p-value = 0.4015
## alternative hypothesis: two.sided
```

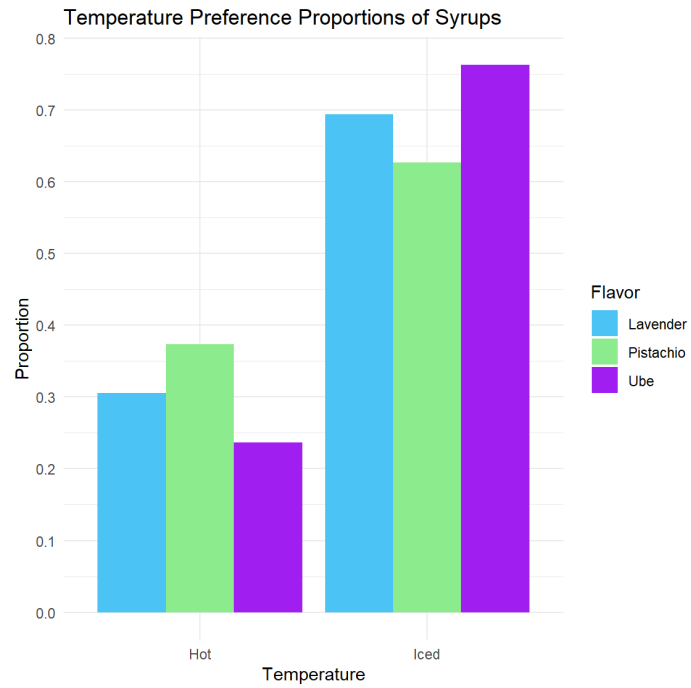
After running the test, a p-value of 0.4015 is received. This means that if milk preference were truly independent of syrup flavor (the null hypothesis), there is a 40.2% chance of observing a result at least as extreme as ours just by random chance. Since this is well above our significance threshold of 0.05, we fail to reject the null hypothesis and conclude that milk preference is not significantly associated with syrup flavor choice. In other words, customers choose their milk independently of their spring syrup choice.

What is the difference in sales of the syrups between Hot vs Iced?

To look at the difference between getting the hot and iced variation of the syrups, first the counts of iced and hot variation of each syrup was collected and put into a 2x3 Matrix. The rows were the temperature and columns were the flavor. Then the matrix of counts and of proportions were both plotted:

```
##        Ube Lavender Pistachio
## Iced 0.764    0.694     0.627
## Hot  0.236    0.306     0.373
```





From the Counts graph on the left, we can see that the Ube has much more order than the other two, however, this can be attributed to being released a month earlier. Looking at the proportions graph, there seems to be a much higher preference in iced drinks, rather than hot drinks with more than 60% of all three syrup's having the iced variation rather than the hot variation. Pistachio appears to have a more even split with 37% of customers ordering their drink hot, while 63% had an iced preference.

To see if there is a relationship between temperature and flavor, we can run another Chi-Square test of independence. The hypothesizes:

- **H_0:** There is no relationship between Temperature choice and syrup choice
- **H_1:** There is a relationship between Temperature choice and syrup choice

Set the significance level:

- **α:** 0.05

The chi-square test assumptions are listed below:

1. Both of the variables, Temperature choice and syrup choice, are categorical variables
2. Since a customer's order does not effect another customer's order, we can safely make the assumption that every observation is independent
3. Each observation does only fall into one cell i.e every observation has one temperature preference and one syrup type and not multiple
4. Random sampling is where things get tricky. Since all of the data was collected in this one coffee shop, it is harder to say that it was random, but for our purposes we only need it for Cor and we don't have control over our customers so there is some randomness.

Looking at our expected values below, we pass the expected values condition since each number is greater than 5:

```
##            Ube Lavender Pistachio
## Iced 145.95868  61.1157  53.92562
## Hot   57.04132  23.8843  21.07438
```

```
##
##  Pearson's Chi-squared test
##
## data:  temp_mat
## X-squared = 5.4192, df = 2, p-value = 0.06656
```

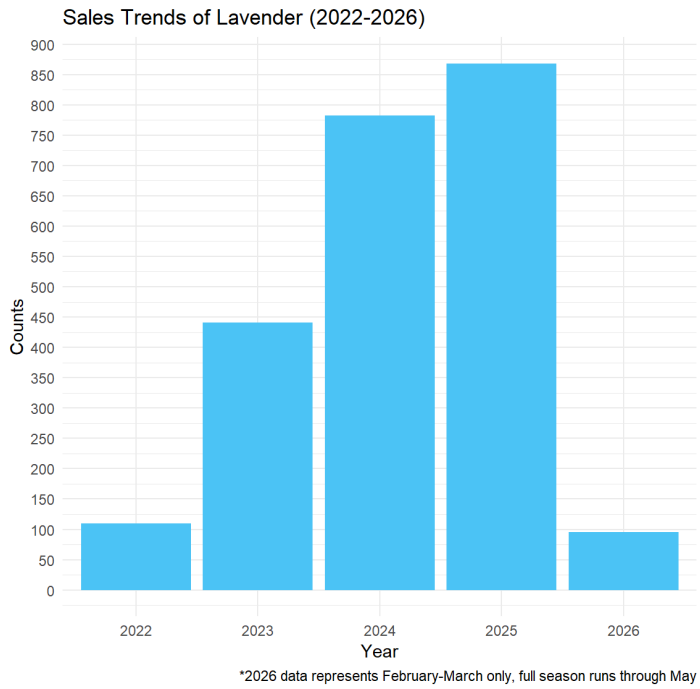
After running the test, we get a X-squared value of 5.4192 and a p-value of 0.06656. This means that under the assumption that the temperature and flavor choices are independent, then there is a 6.66% chance of getting a value as extreme as this on chance alone. While this does seem to be very low, we set our significance level to 0.05, and since our p-value is greater than our significance level, we don't have enough evidence to say that there is a relationship between the temperature and flavor choice, so fail to reject null.

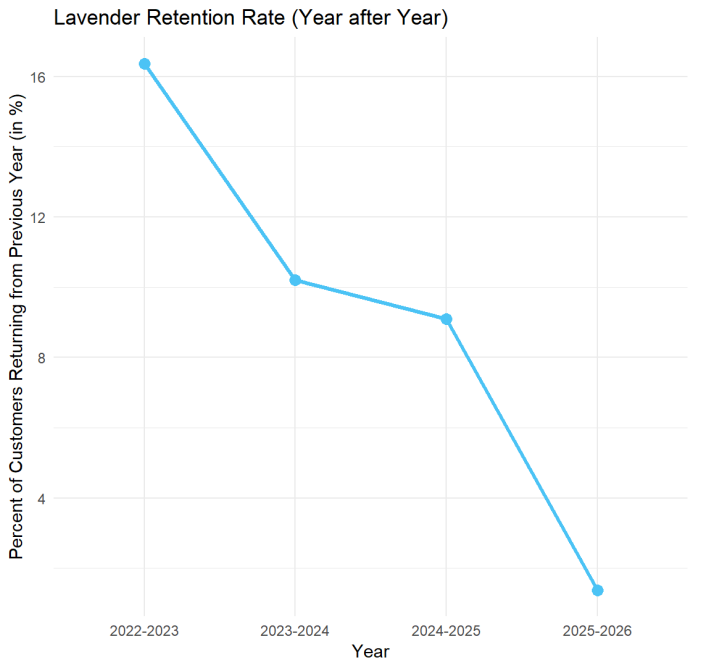
How many people from previous years are repeat drinkers of Lavender?

To see how many people from previous years continued to drink Lavender, all of the non-blank PAN suffix values were taken for every year since 2022, the year it came out. There was a format change for PAN suffix numbers in 2022 and 2023 where some numbers had less than 4 digits. To fix this, a 0 was added in front of every non 4 digit PAN number. After cleaning the data, the function intersect was used to see how many people from the previous year returned. Here are the results:

Lavender Returners	Customer Retained	Customer Retaination Rate (%)
2022 - 2023	18	16.36 %
2023 - 2024	45	10.2 %
2024 - 2025	64	9.1 %
2025 - 2026	11	1.38 %
2022 - 2025	1	0.91 %
2022 - 2026	0	0 %

After gathering the retention rate data and gathering the counts of Lavender orders from the year it was released to now, here are the plots:





The graph on the left shows the number of Lavender orders from when it was released, while the graph on the right shows the rate of Lavender returners over the years. From the sales graph, a clear increase in sales can be shown aside from 2026. Although the semester isn't quite over yet, that is a very overwhelming gap between 2025 and 2026. A possible explanation for this tremendous gap, could be releasing Pistachio alongside Lavender. From the cumulative sales graph it was shown that Lavender and Pistachio were very close in sales and from the milk and temperature preferences charts, it was shown that customers had similar preferences for both syrups as both had close temperature preferences and in the preferences, both syrups had about the same proportion for all the syrups with the exception of Soy milk which had a bigger Lavender preference and Coconut milk which lacked any Lavender presence, but in both of these cases there were only a few drinks ordered with these syrups, and in the Syrup Sales by Drink graph, the preference for the Latte and Matcha, the two highest selling drinks, were very close.

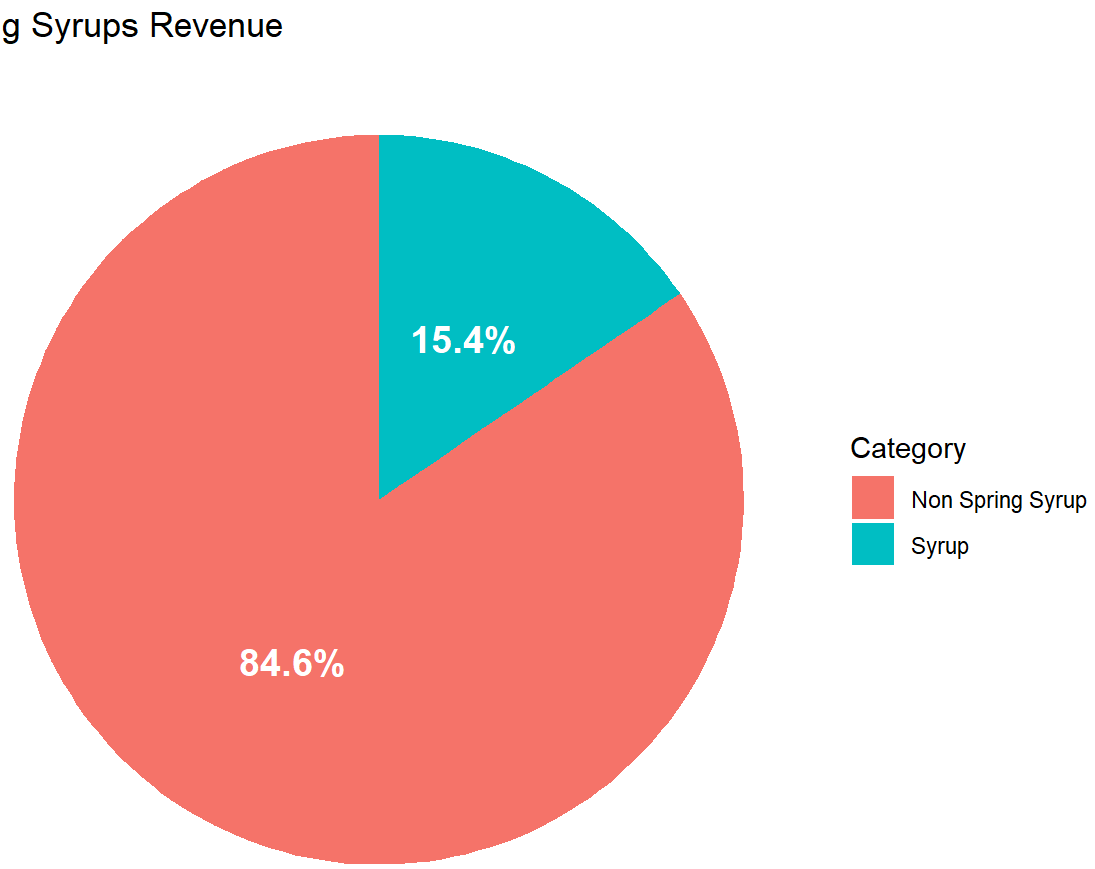
Even though the Lavender has been ordered more frequently over the years, the retention of customers have been on a decline. There are many explanations to this such as the Lavender customers from previous years simply graduating and not coming back to Cor or perhaps the customers have switched to try another syrup. To test how many customers switched to Ube and Pistachio, the PAN suffixes of all the orders of Pistachio and Ube in 2026 was gathered and then the intersection of both of those syrups and Lavender Pan Suffixes were found. Here are the results:

Syrup	Switchers	Switch rate from Lavender customers in 2025
Ube	27	3.38%
Pistachio	8	1.00%
Lavender	11	1.38%

The table shows that all the customers that ordered Lavender in 2025, about 3.38% of them switched to buying Ube, while 1.00% switched to buying Pistachio, and only 1.38% of customers continued to buy Lavender. Ube appears to be the syrup that 2025 Lavender customers prefer over the Pistachio and Lavender, but this could be due to those customers not being frequent customers and just buying a small treat for themselves. We can test this by seeing if the customers that buy spring syrups are a different kind customer.

Are the customers that buy spring syrups a differnt kind of customer?

What does it mean to be a different kind of customer? In this instance, a different customer is just a customer that spends more on average. To see if a customer spends more than average, first it is important to know what percent of the total revenue of Cor is due to the spring syrup drinks:



15.4% of the total revenue of Cor is generated from the spring syrups, which is a sizable proportion. This shows that those syrups do bring in a lot of revenue, which could possible mean either: the customers that buy those drinks are heavy spenders on average or it could be due to those syrups being slightly more expensive than the staple syrups. By running a one-way ANOVA test, the transaction values between syrup and non-syrup orders can be compared.

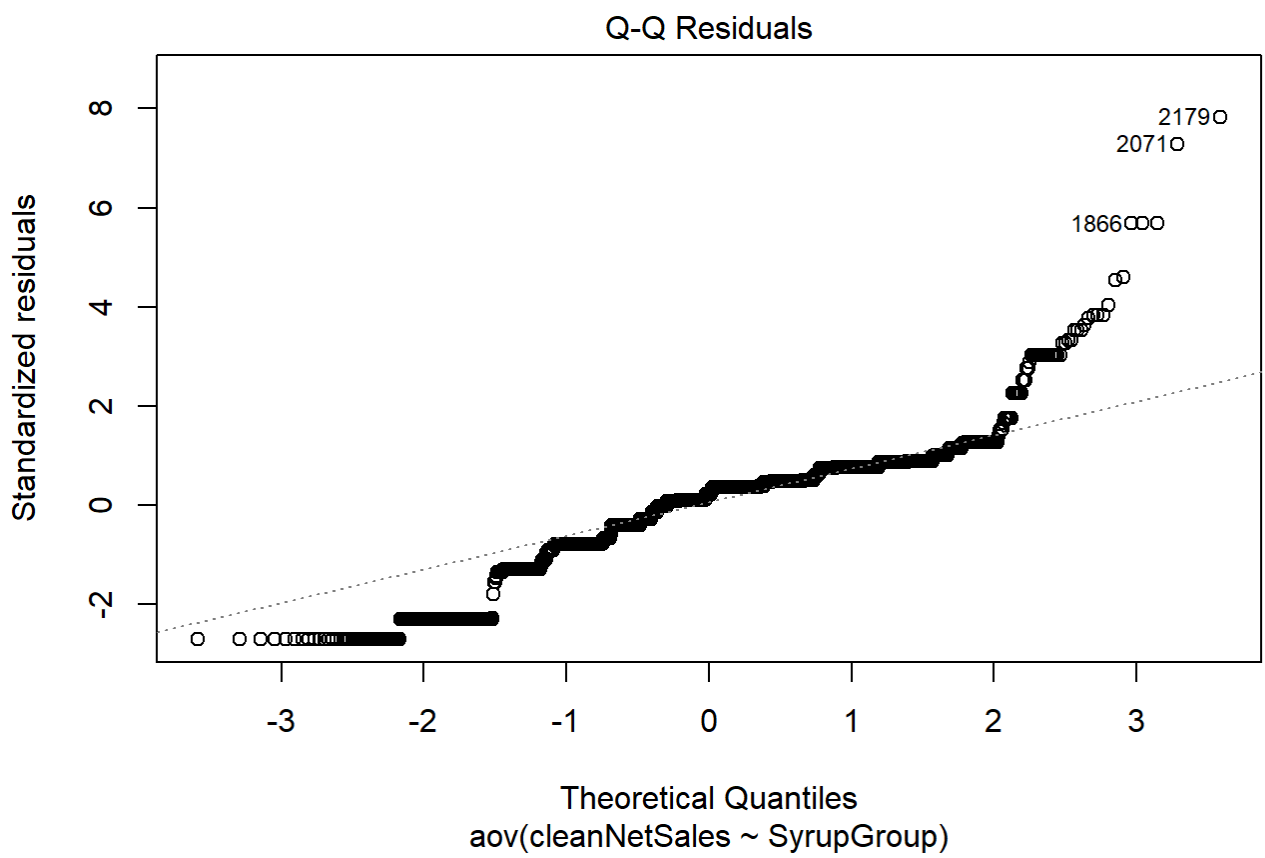
- **H_0:** The mean transaction value is the same for syrup and non-syrup orders
- **H_1:** The mean transaction value is different between syrup and non-syrup

Significance Level:
- **α:** 0.05
Before we can start the ANOVA test, we must see if we pass the checks:
1. Homogeneity of Variance:
To see if the variances of the Net Sales and Syrups are similar, Levene's test is run:

```
## Levene's Test for Homogeneity of Variance (center = median)
##         Df F value Pr(>F)
## group    1  2.3816 0.1229
##       2997
```

The p-value of 0.1229 signifies that there isn't convincing evidence to suggest that the variances aren't similar. So we proceed with using ANOVA.

2. Independence of Observations:
   Since each transaction is independent of others (the previous transaction doesn't affect the next transaction), then it can be assumed that the observations are independent.
3. Normality of Residuals:
   To see if the residuals are normally distributed, we graph the residuals on a plot and see if they follow a linear pattern.



The plot reveals that the residuals don't follow a linear path so it could be assumed that the residuals aren't normal. However, due to our large sample size of 3002 observations, this test failing is mitigated, but to be sure we will compare our ANOVA results with the Kruskal-Wallis test to see if both results reach the same conclusion.

```
##               Df Sum Sq Mean Sq F value   Pr(>F)
## SyrupGroup     1    218  217.80   55.95 9.73e-14 ***
## Residuals   2997  11668    3.89
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 3 observations deleted due to missingness
```

Running the ANOVA test returns a p-value of 9.73e-14. This value is infinitesimally small, so small that it is very unlikely that the **H_0** will fail to reject, but since we failed one of our assumptions we run the Kruskal-Wallis Test. This is the ANOVA test compares the actual means between the groups and assumes the data is normally distributed, while the Kruskal-Wallis test only compares the ranks of the values between groups and doesn't make the assumption about the distribution's being normally distributed:

```
##
##  Kruskal-Wallis rank sum test
##
## data:  cleanNetSales by SyrupGroup
## Kruskal-Wallis chi-squared = 203.24, df = 1, p-value < 2.2e-16
```

The Kruskal-Wallis test returned a value 2.2e-16, which is equally a small number. Since both tests agree on the same conclusion and are significantly smaller than our p-value of 0.05, it is clear that we have convincing evidence to suggest that the mean transaction value is different between syrup and non-syrup. So we reject **H_0**. This conclusion makes a lot of sense since the spring syrups are about $0.25 more expensive than the staple syrups (which aren't counted in the spring syrups group). So the mean transactional value of spring syrup drinks will always be more expensive than drinks without Ube, Lavender, or Pistachio.

But to be sure if this value is explained by the customers rather than the extra cost of the syrups, we can gather all the customers that have never gotten a spring syrup and find their mean transactional value of every drink they've gotten and compare it to the customers that have gotten a spring syrup and seeing their transactional value of drinks without the spring syrups. If the mean transactional value of the customers that get the spring syrups is higher than the customers that have never gotten the spring syrups, then this will show us whether the customers that do get spring syrups are heavy spenders or if they get Ube, Lavender, and/or Pistachio as a sweet treat.

Customer Type	Average spending
Mean value of Syrup Customer Without Syrup transactions	$ 3.93
Mean value of Non-Syrup Customer transactions	$ 4.54

The table above reveals that the average transactional value of order from customers that order Ube, Lavender, or Pistachio that don't contain any is smaller than the mean value of customers that don't order the syrups. This

shows that the customer that buy the spring syrups aren't heavy spenders but it is due to other reasons, one possible reason being that they buy these syrups simply as a sweet treat for themselves or every once in a while.

To fully see our customer base, we can gather all of the PANs of the customers that order Ube, Lavender, and Pistachio and see how often each of them visit. Here is the summary of the data:

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   1.000   1.000   2.000   3.951   5.000  34.000
```

Using the median as a point of center since it is not influenced by outliers, it appears that the average customer will buy about 2 drinks of the spring syrups. This supports the conclusion that the customers buy the spring syrups as a treat every once in a while. However, the max is 34 drinks of syrups which is a huge outlier. What this tells us is that there are very loyal customers to these drinks. By looking for only visits that are above the 75th percentile of Syrup buyers (5) and dividing it by the total number of syrup buyers, we find that about 22% of customers are very loyal to the spring syrups. In conclusion the customers that buy spring syrups tend to fall into 2 categories: either customers that want a small treat every once in a while or customers that are big fans of the syrups.

What predicts whether a customer orders Ube, Lavender, or Pistachio?

To see what variables predicts the spring syrup a customer orders we can build a simple multinomial logistic regression model. The variables used in this model are if the drink is iced, the hour of the day, and the type of milk. Month is excluded since Ube had started a month prior to Lavender and Pistachio. After building the model, here are the results:

```
##           (Intercept)     IsIced       hour MilkCoconut   MilkOat   MilkSoy
## Lavender    0.9416429 0.38350821 0.79878718   0.0000000 0.1986094 0.4393628
## Pistachio   0.1507853 0.06593004 0.09871881   0.2434433 0.1228078 0.0000000
##           MilkWhole
## Lavender  0.3535017
## Pistachio 0.1385488
```

This model is comparing Lavender and Pistachio to Ube, so we know if someone is more likely to get Lavender or Pistachio over Ube. Looking at the p-values of the variables, we see that none of the variables are very statistically significant since they are all above 0.05 (except for Coconut Milk, but this is most certainly due to a smaller sample size). The variables are the closest are whether or not the drink is iced and the hour of the day, both for Pistachio, this makes sense because in the sales per hour density plot, we saw that there was lot more Pistachio sales in the latter half of the day compared to Ube selling more in the beginning of the day and Pistachio's hot and cold drinks was more split compared to Ube which had an overwhelmingly cold preference. Taking a look at the coefficients:

```
## Call:
## multinom(formula = Flavor ~ IsIced + hour + Milk, data = syrupdf)
##
## Coefficients:
##           (Intercept)     IsIced        hour MilkCoconut    MilkOat    MilkSoy
## Lavender  -0.06230646 -0.2258912 -0.01605012  -16.467802 -0.6482574   1.015655
## Pistachio -1.26736981 -0.4859424  0.10698252   -1.399094 -0.7867100 -12.203205
##            MilkWhole
## Lavender  -0.4546247
## Pistachio -0.7363311
##
## Std. Errors:
##           (Intercept)    IsIced       hour  MilkCoconut   MilkOat      MilkSoy
## Lavender    0.8511218 0.2592129 0.06296215 1.718740e-07 0.5042751 1.313456e+00
## Pistachio   0.8821006 0.2642572 0.06479482 1.199474e+00 0.5098261 6.550233e-06
##           MilkWhole
## Lavender  0.4899931
## Pistachio 0.4971123
##
## Residual Deviance: 783.6583
## AIC: 811.6583
```

The coefficients show mostly a weaker relationship for all the milk based variables, which makes sense since as we didn't have a big sample size for the milks besides oat milk and whole milk and our fisher test showed that there wasn't a significant relationship between the syrups and the milks. Whole milk for both Lavender and Pistachio are both less than -0.6 which shows that customers have a stronger preference for Ube when they want Oat milk or Whole milk.

For the Iced variable, the coefficients are much weaker since they are greater than -0.5 but less than 0. However, the negative number indicates that if the drink is iced than the customer prefers Ube, but Lavender is able to compete more compared to the pistachio since it's coefficient is about -0.226 compared to Pistachio's -0.486.

The weakest variable by far is the hour variable where both variables are in between -0.1 and 0.2 which shows a very weak relationship. However, the Pistachio coefficient is positive, which means that as the day goes on, customers tend to order Pistachio, which saw in the hour density plot, that the Pistachio sales peaked in the latter half of the day.

Overall, after looking at the results of the multinomial logistic regression table, the results tell us that it is most likely just the preference of the customer's choice of the syrup rather than the type of milk, temperature, and hour of the day that affect the syrup the customer gets in their drink.

Conclusion

Ube, Lavender, and Pistachio are Cor Coffee's newest syrups that have had a well reception since they came out. This analysis was created to see what drove these syrups to be well liked. Overall, Ube was consistently dominating in every category compared to the Lavender and Pistachio. Ube was the most preferred and widely used syrup in the drinks. The Lavender being second most preferred and widely used, however, the Pistachio was very close behind in both preference and usage in drinks. Ube had the most sales and stayed on top all throughout. Lavender had a weaker start but continued to be a steady favorite and the Pistachio had a strong start but then slowed down to be less than Lavender. Time wise, All three tended to sell a good amount during the first

two hours and after 12 PM mass, which is aligned with the morning and after Mass rushes. Ube dominated in the mornings, while Pistachio dominated in the afternoon. Lavender remained to be pretty consistent throughout the day. It was found out that milk type didn't have a significant affect on syrup choice, however Ube was chosen the most in every category (except soy, but that is most likely due to a lack of soy orders). Lavender and Pistachio were very comparable in milk choice since both were head to head in most milk excluding coconut and soy which didn't have a big sample size. For the temperature, customer overwhelmingly preferred iced variations of the syrups compared to their hot counterparts. The Pistachio, however did have the most even split, which makes sense since it is the syrup that is used mostly in hotter drinks.

Since Lavender came out in 2022, it has been a popular syrup choice and has grew at a steady rate. However, there was a big drop off in 2026 and the retention rate showed a big decrease in the same customers buying Lavender through the years. This could be due to the customers that are loyal Lavender fans, graduating and not ordering it again or those Lavender customers switching to another syrup. We found that about 3.38% of Lavender customers in 2025 switched completely in buying Ube and 1.00% to buying Pistachio while only 1.38% stayed with Lavender. This doesn't necessarily mean that customers don't like Lavender, but rather customers could be trying other syrups.

Looking at the revenue of Cor, we saw that the spring syrup drinks made up about 15.4% of the total revenue. The customers that bought the spring syrups had a higher mean transactional value than customers that didn't buy the spring syrups which was shown through our ANOVA test and Kruskal-Wallis test, but this was explained by the syrups being more expensive than everything else. By comparing the mean value of Syrup Customer Without Syrup's transactions and non-syrup customer's transactions alongside seeing the number of syrup orders from the customers that buy spring syrups, we found that there are two types of customers: those that want a sweet treat every once in a while and those that are big fans of the new syrups.

The model that we built showed that there wasn't a single variable that could predict the type of syrup a customer would order, but instead it is most likely personal preference that indicates whether a customer chooses Ube, Lavender, and Pistachio. The coefficients did reinforce our previous findings that Ube dominated the Iced temperature preference, Pistachio sells better in the latter half of the day, and finally that Milk choice didn't predict syrup choice.

There were limitations in this report that was stated previously, but will be restated such as Ube having a longer sales window than the Lavender and Pistachio since it was released in February 17th while the Lavender and Pistachio were both released March 2nd. Lavender is more well-known since it has been released since it has been released as a seasonal syrup in previous years. There were small sample sizes for alternative milk types except for Oat milk. There wasn't a perfect method to select how customers were identified, the best option being choosing the PAN suffix but this option lost about 6% of the data and the customers could be paying with different cards so it doesn't do the best job at identifying customers. Finally, this is incomplete since the semester is not over yet, however the results were pretty absolute and there is less than a month left of sales so it is not very likely that the results will change significantly.

Some recommendations for Cor Coffee:

1. Since Pistachio and Lavender were very close together in sales, milk preference, and temperature preference, customers seem to go to these syrups divide between these syrups. Instead of selling them together, sales might increase if one is sold after the other is done selling.

2. Ube dominated all of the syrups by a noticeable margin. The sales of Ube showed that even a month past it's release date, it was still very popular, growing at an impressive rate. Because of this, keeping Ube as a staple syrup is something worth considering.

3. From our results we found that there are most likely 2 types of customers at Cor: those that get the spring syrups as a sweet treat and those that are very big fans of the spring syrups. Currently Cor has a punch card system that gives one punch per drink bought. To incentivize both groups to buy more syrup drinks, Cor could offer a limited time 2 punches per special syrup bought on the punch card or another punch card system for specifically the spring syrups. Even if this does end up temporarily lowering revenue, it would bring in more customers for the future and the purpose of Cor is helping college students to encounter Christ and become fully alive in him and this would bring more customers through the door allowing for this mission to be better done.